

StruMamba3D: Exploring Structural Mamba for Self-supervised Point Cloud Representation Learning

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Motivation & Key Challenges

- Mamba-based models show promise for point cloud learning
- Two main challenges:
 - - Destroyed 3D point adjacency during SSM processing
 - - Poor long-sequence memory retention for longer inputs
- Need to preserve spatial dependencies and adapt to varying sequence lengths

StruMamba3D: Paradigm Comparison

- Transformer: Quadratic complexity, models dependencies via attention
- Mamba: Linear complexity, but loses spatial adjacency and long-sequence memory
- StruMamba3D: Maintains spatial dependencies and adapts to sequence length
- Outperforms Transformer and Mamba-based methods

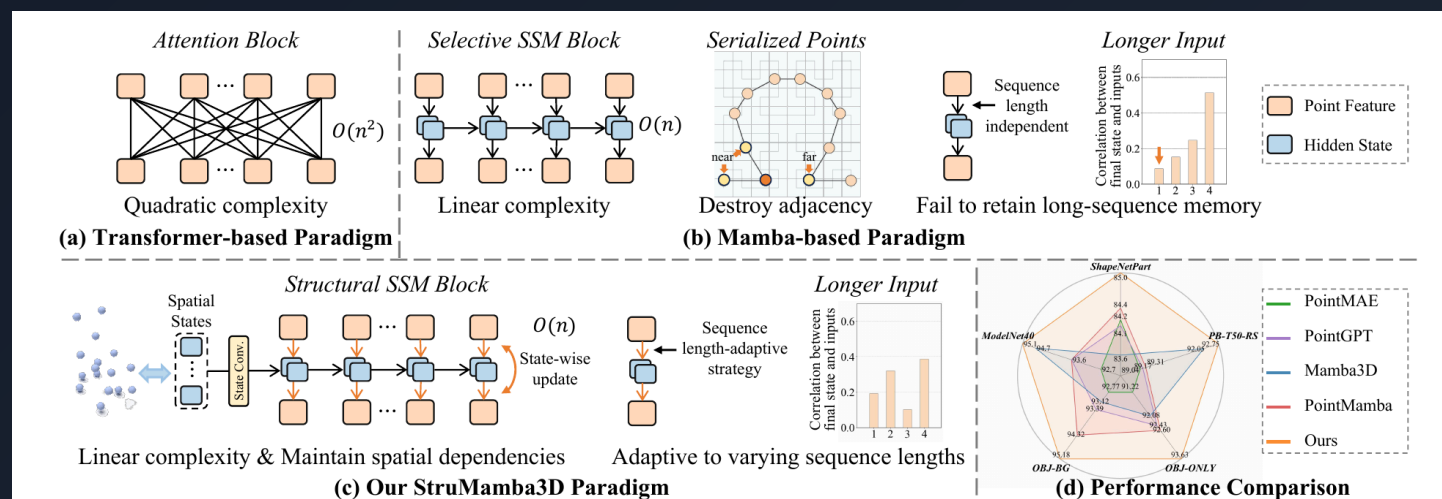


Figure 1. (a): Transformer-based paradigm uses the attention block with quadratic complexity to model the dependencies between points. (b): Mamba-based paradigm uses the selective SSM with linear complexity. However, serialized points destroy the adjacency of 3D points, and the pre-trained selection mechanism fails to retain long-sequence memory. (c): Our StruMamba3D paradigm uses the structural SSM to maintain the spatial dependencies among points and the sequence length-adaptive strategy to retain long-sequence memory. (d): Our StruMamba3D

StruMamba3D Architecture Overview

- Spatial states initialized as proxies for local structure
- Structural SSM blocks capture spatial dependencies
- Lightweight convolution introduces local interactions
- Bidirectional scanning for enhanced information flow

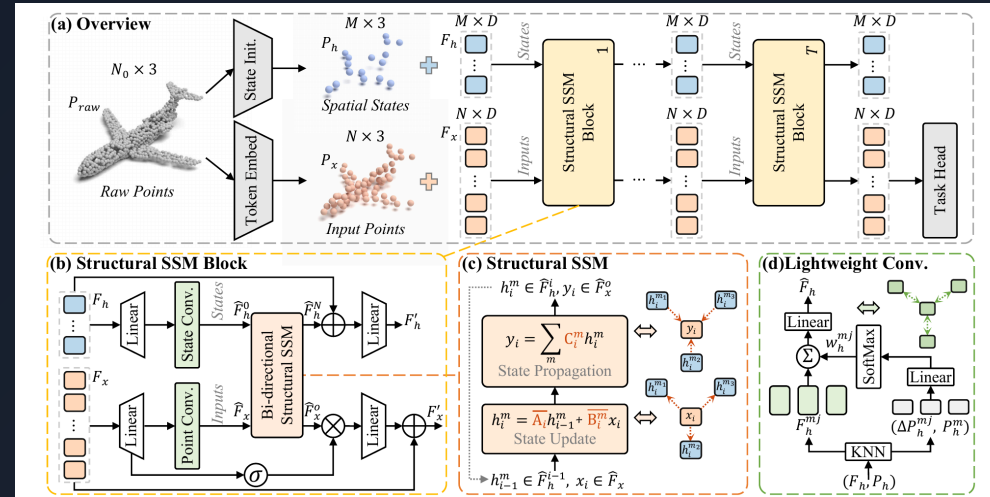


Figure 2. Overview of StruMamba3D. (a): Architecture of StruMamba3D. (b): Details of Structural SSM Block. We employ structural SSM to capture dependencies between spatial states and input points. Additionally, we integrate a lightweight convolution module to introduce local interactions among spatial states or input points. (c): Details of Structural SSM. (d): Details of Lightweight Convolution.

Structural SSM Block: Key Components

- Spatial state initialization assigns positions to latent states
- State-wise update models spatial dependencies between points and states
- Bidirectional SSM for richer feature propagation
- Lightweight convolution enhances inter-state interactions

Sequence Length-Adaptive Strategy

- Adaptive state update mechanism adjusts update frequency by sequence length
- Spatial state consistency loss enforces robustness to varying input lengths
- Improves long-sequence memory retention in downstream tasks

Pre-training Framework

- Masked Point Modeling (MPM) as main pre-training task
- Spatial state consistency loss aligns student and teacher state features
- Teacher updated via exponential moving average

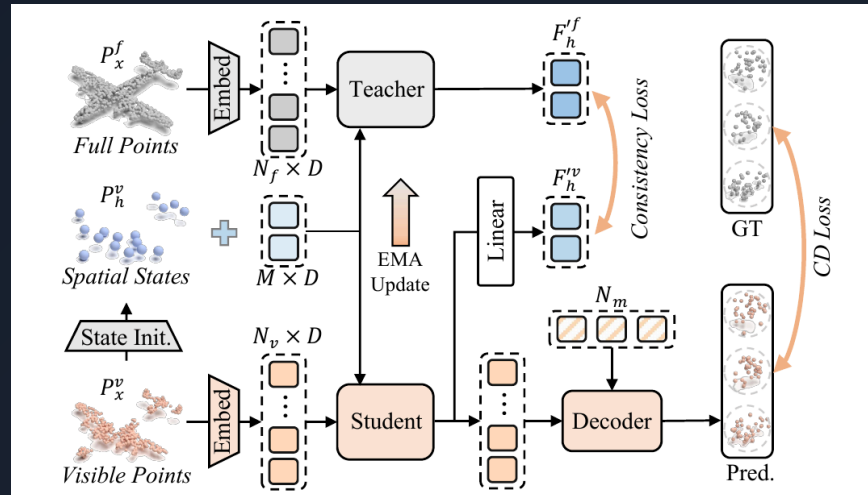


Figure 3. Overview of the pre-training framework. We pretrain the StruMamba3D (student mode1) using the MPM task and the spatia1 state consistency loss. The teacher mode1 has the same structure as the student mode1, but its parameters are updated using the exponentia1 moving average strategy.

Shape Classification Results

- StruMamba3D achieves SOTA on ScanObjectNN and ModelNet40
- Outperforms Transformer and Mamba-based methods
- Significant gains in challenging splits and without voting

Method	Backbone	Param. (M)	FLOPs (G)	ScanObjectNN			MODELNET40	
				OBJ-BG	OBJ-ONLY	PB-T50-RS	w/o Voting	w/ Voting
Supervised learning only								
PointNet [27]	-	3.5	0.5	73.3	79.2	68.0	89.2	-
PointNet++ [28]	-	1.5	1.7	82.3	84.3	77.9	90.7	-
PointCNN [19]	-	0.6	-	86.1	85.5	78.5	92.2	-
DGCNN [35]	-	1.8	2.4	82.8	86.2	78.1	92.9	-
PRANet [8]	-	2.3	-	-	-	81.0	93.7	-
PointNeXt [30]	-	1.4	3.6	-	-	87.7	94.0	-
PointMLP [24]	-	12.6	31.4	-	-	85.4	94.5	-
DeLA [2]	-	5.3	1.5	-	-	88.6	94.0	-
PCM [47]	-	34.2	45.0	-	-	88.1	93.4	-
Pre-training using single-modal information								
PointBERT [42]	Transformer	22.1	4.8	87.43	88.12	83.07	92.7	93.2
MaskPoint [22]	Transformer	22.1	4.8	89.30	88.10	84.30	-	93.8
PointM2AE [44]	Transformer	12.7	7.9	91.22	88.81	86.43	92.9	93.4
PointMAE [†] [25]	Transformer	22.1	4.8	92.77	91.22	89.04	92.7	93.8
PointGPT-S [†] [3]	Transformer	29.2	5.7	93.39	92.43	89.17	93.3	94.0
PointMamba [†] [21]	Mamba	12.3	3.1	94.32	92.60	89.31	93.6	94.1
Mamba3D [†] [15]	Mamba	16.9	3.9	93.12	92.08	92.05	94.7	95.1
Ours [†]	Structural SSM	15.8	4.0	95.18	93.63	92.75	95.1	95.4
Pre-training using cross-modal information								
ACT [†] [7]	Transformer	22.1	4.8	93.29	91.91	88.21	93.7	94.0
Joint-MAE [14]	Transformer	22.1	-	90.94	88.86	86.07	-	94.0
I2P-MAE [†] [46]	Transformer	15.3	-	94.15	91.57	90.11	93.7	94.1
ReCon [†] [29]	Transformer	43.6	5.3	95.18	93.29	90.63	94.5	94.7

Table 1. Shape Classification on ScanObjectNN and ModelNet40 Datasets. For ScanObjectNN, we report the classification accuracy(%) over the three subsets: OBJ-BG, OBJ-ONLY, and PB-T50-RS. And † indicates using simple rotational augmentation [7] for fine-tuning.

Part Segmentation Results

- StruMamba3D achieves highest mIoU among single-scale models
- Outperforms Mamba3D and PointMamba on ShapeNetPart
- Efficient structure modeling without serialization

Method	5-way		10-way	
	10-shot	20-shot	10-shot	20-shot
PointBERT [42]	94.6 $\pm$ 3.6	93.9 $\pm$ 3.1	86.4 $\pm$ 5.4	91.3 $\pm$ 4.6
MaskPoint [22]	95.0 $\pm$ 3.7	97.2 $\pm$ 1.7	91.4 $\pm$ 4.0	92.7 $\pm$ 5.1
PointMAE [25]	96.3 $\pm$ 2.5	97.8 $\pm$ 1.8	92.6 $\pm$ 4.1	93.4 $\pm$ 3.5
PointM2AE [44]	96.8 $\pm$ 1.8	98.3 $\pm$ 1.4	92.3 $\pm$ 4.5	95.0 $\pm$ 3.0
PointGPT-S [3]	96.8 $\pm$ 2.0	98.6 $\pm$ 1.1	92.6 $\pm$ 4.6	95.2 $\pm$ 3.4
PointMamba [21]	96.9 $\pm$ 2.0	99.0 $\pm$ 1.1	93.0 $\pm$ 4.4	95.6 $\pm$ 3.2
Mamba3D [15]	96.4 $\pm$ 2.2	98.2 $\pm$ 1.2	92.4 $\pm$ 4.1	95.2 $\pm$ 2.9
Ours	97.5$\pm$2.3	99.1$\pm$1.4	93.5$\pm$3.7	96.1$\pm$3.5

Table 2. Part Segmentation on ShapeNetPart. We report the mean IoU for all classes (mIoU) and all instances (mIoU).

Large-scale 3D Scene Understanding

- Evaluated on ScanNetV2 for 3D object detection
- StruMamba3D achieves highest mAP25 and mAP50
- Demonstrates scalability to large point clouds

Backbone	Serialization	Overall Accuracy		mIoU _c
		ScanNN	MN40	SNPart
Mamba	✗	88.24	92.50	82.08
Mamba	Z-order	89.28	93.44	83.89
Mamba	Hilbert	89.87	93.68	84.07
StruMamba3D	✗	<u>92.75</u>	95.06	<u>84.96</u>
StruMamba3D	Z-order	92.47	<u>94.98</u>	84.86
StruMamba3D	Hilbert	92.81	94.89	85.12

Tab1e 9. Quantitative comparison on the ScanNetV2 dataset in terms of mAP25 and mAP50.

Computational Efficiency

- StruMamba3D has linear complexity via hardware-aware scan
- Lower FLOPs than Transformer-based PointMAE for long sequences
- Faster inference and higher FPS, especially with longer inputs

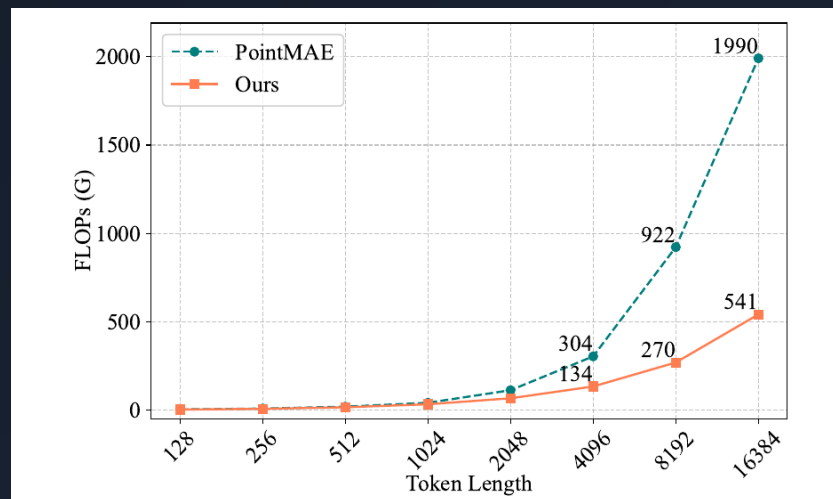


Figure 4. FLOPs comparison with PointMAE.

Inference Time and FPS Comparison

- StruMamba3D outperforms PointGPT and Mamba-based baselines in speed
- Maintains efficiency as input length increases

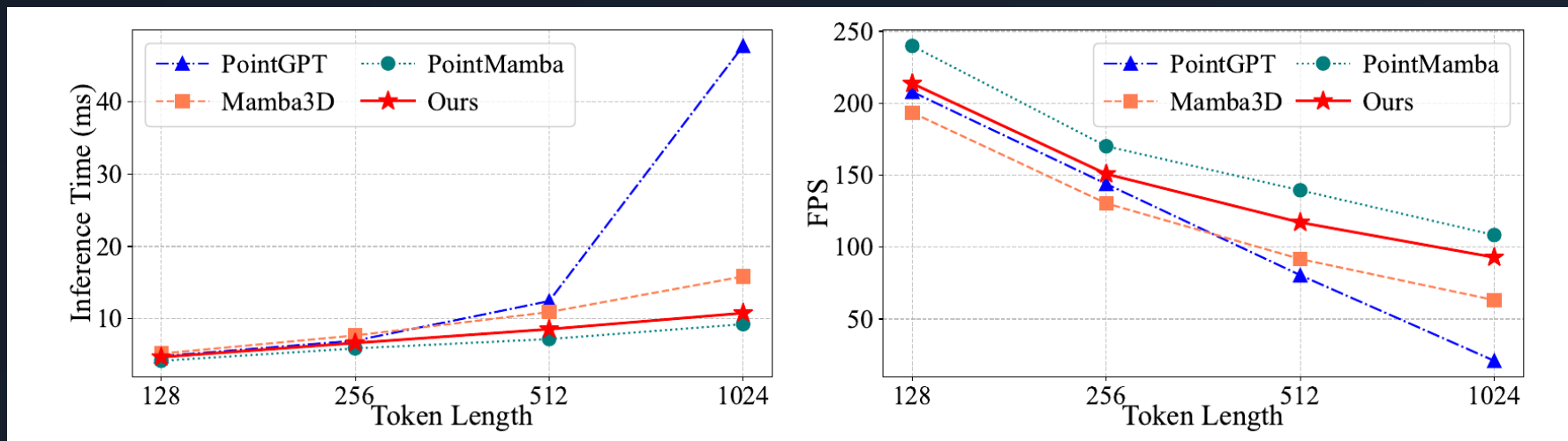


Figure 5. Inference time and FPS comparison.

Ablation: Structural SSM Block

- Structural SSM block yields large accuracy gains
- Lightweight convolution and bidirectional scanning further improve results
- Spatial state modeling is crucial for performance

Method	Overall Accuracy		mIoU _c
	ScanNN	MN40	SNPart
baseline	88.24	92.50	82.08
w/ Structural SSM	91.78	93.92	84.15
w/ Lightweight Conv. for h	92.22	94.65	84.62
w/ Lightweight Conv. for x	92.40	94.81	84.77
w/ Bidirectional Scanning	92.75	95.06	84.96

Table 5. Ablation on Structural SSM Block. The baseline uses the standard Mamba block to replace the Structural SSM Block.

Ablation: Designed Modules

- Structural SSM block and sequence length-adaptive strategy both boost accuracy
- Combining both achieves optimal results across tasks

Structural SSM Block	SLAS	Overall Accuracy		mIoU _c
		ScanNN	MN40	SNPart
$\times$	$\times$	87.23	91.86	81.56
$\checkmark$	$\times$	92.09	94.45	84.49
$\checkmark$	$\checkmark$	92.75	95.06	84.96

Table 4. Effect of the designed modules. ‘SLAS’ indicates the sequence length-adaptive strategy used in pre-training phase.

Ablation: Structural SSM Details

- State-wise update strategy is essential for structural modeling
- Combining point features and spatial relationships yields best results

$\phi(x)$	SS	MLP(ΔP)	Overall Accuracy		mIoU _c
			ScanNN	MN40	SNPart
✓	✗	✗	90.94	93.84	83.62
✓	✓	✗	91.33	94.12	83.87
✓	✓	✓	92.75	95.06	84.96
✗	✓	✓	91.12	94.25	84.02

Table 6. Ablation on Structural SSM. 'SS' represents the use of spatial states in the SSM. 'a' denotes the SSM parameters depending on the point feature, while 'MLP(AP)' represents the SSM parameters depending on the spatial relationship.

Ablation: Sequence Length-Adaptive Strategy

- Adaptive state update and consistency loss are complementary
- Best performance when both are used together
- Model is robust to varying input lengths

ASUM	$\mathcal{L}_{ssc}$	Overall Accuracy		
		ScanNN	MN40	SNPart
$\times$	$\times$	92.09	94.45	84.49
$\checkmark$	$\times$	92.26	94.57	84.56
$\checkmark$	$\lambda = 1$	92.47	94.94	84.81
$\checkmark$	$\lambda = 2$	92.75	95.06	84.96
$\checkmark$	$\lambda = 5$	92.57	94.89	84.77
$\times$	$\lambda = 2$	92.16	94.65	84.56

Table 7. Ablation on sequence length-adaptive strategy. ‘ASUM’ indicates the adaptive state update mechanism, and ‘ λ ’ indicates the weight of the spatial state consistency loss $\mathcal{L}_{ssc}$.

Ablation: Point Convolution

- Number of neighbors in lightweight convolution impacts performance
- Optimal: $k=4$ for spatial states, $k=8$ for input points
- Spatial states more sensitive to neighbor count

Method	ScanObjectNN	ModelNet40	ShapeNetPart
	mOA	mOA	mIoU _c
w/o Pretraining	91.33	93.68	83.96
MPM Pretraining	92.09	94.45	84.49
Our Pretraining	92.75	95.06	84.96

Table 12. Ablation on the point convolution. “ k ” denotes the number of k -nearest neighbors in the lightweight convolution,

Detailed Part Segmentation Results

- StruMamba3D achieves best mean IoU in most categories
- Spatial states enhance local structure modeling
- Superior segmentation of complex shapes

Method	mIoU _c	mIoU _i	airplane	bag	cap	car	chair	earphone	guitar
MaskPoint [22]	<u>84.6</u>	86.0	84.2	85.6	88.1	80.3	91.2	79.5	91.9
PointBERT [42]	84.1	85.6	84.3	84.8	88.0	79.8	91.0	81.7	91.6
PointMAE [42]	84.2	<u>86.1</u>	84.3	85.0	88.3	80.5	<u>91.3</u>	78.5	<u>92.1</u>
PointGPT-S* [3]	84.0	85.9	85.1	<u>86.1</u>	<u>88.8</u>	80.2	91.2	78.8	<u>91.7</u>
PointMamba* [21]	83.8	85.9	84.5	84.7	87.8	80.2	91.2	78.7	91.9
Ours	85.0	86.7	<u>84.8</u>	87.4	89.2	81.3	91.6	<u>80.8</u>	92.2
Method	knife	lamp	laptop	motorbike	mug	pistol	rocket	skateboard	table
MaskPoint [22]	87.8	86.2	95.3	<u>76.9</u>	<u>95.0</u>	85.3	64.4	<u>76.9</u>	81.8
PointBERT [42]	87.9	85.2	95.6	<u>75.6</u>	94.7	84.3	63.4	76.3	81.5
PointMAE [42]	87.4	<u>86.1</u>	96.1	75.2	94.6	84.7	<u>63.5</u>	77.1	82.4
PointGPT-S* [3]	87.0	84.5	<u>95.9</u>	74.2	<u>95.0</u>	83.8	62.6	76.8	81.6
PointMamba* [21]	87.3	85.1	95.8	74.0	94.6	84.7	61.6	75.9	82.4
Ours	88.3	86.0	95.8	78.1	95.5	85.8	<u>63.5</u>	<u>76.9</u>	<u>82.2</u>

Table 8. Detailed Results of Part Segmentation on ShapeNetPart Dataset. We report the mean IoU for each category. * indicates results reproduced using the official code.

Visualization: Spatial State Correlation

- Different spatial states focus on distinct regions
- Structural SSM effectively captures local structure

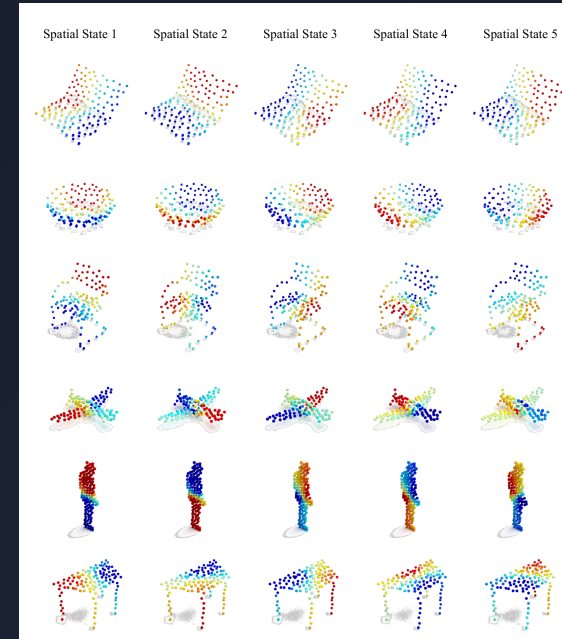


Figure 6. Visualization of correlation between the output point and spatial state features produced by the structural SSM. For each point cloud, we select five spatial states that are widely spaced for visualization.

Visualization: Masked Point Modeling

- StruMamba3D reconstructs masked regions accurately
- Handles complex geometric patterns with high fidelity



Figure 7. Visualization of reconstructed masked regions on ShapeNet. Full points represent the raw point cloud, while visible points correspond to the input points with 60% masked. The reconstruction results consist of the predicted points combined with visible points.

Visualization: Part Segmentation

- Accurate segmentation of complex structures
- Corrects some annotation errors in ShapeNetPart
- High mIoU and robust to noisy labels

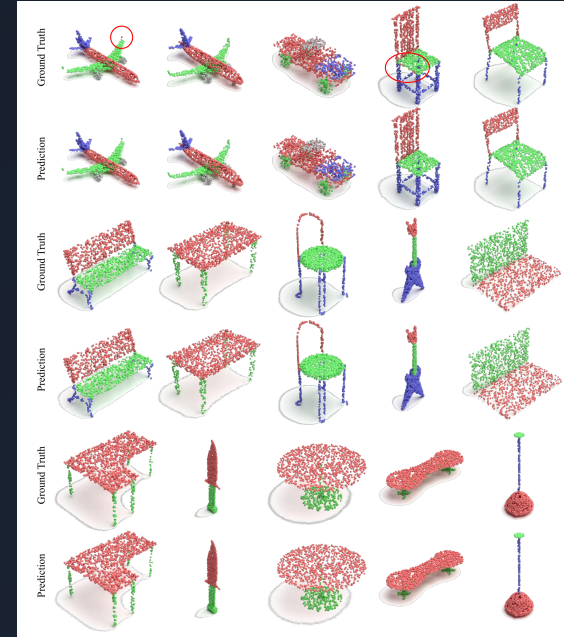


Figure 8. Visualization of part segmentation of our Structural Mamba on ShapeNetPart. Different colors represent different parts. The red circles highlight points with obvious annotation errors.

Conclusion

- StruMamba3D addresses adjacency distortion and long-sequence memory issues
- Structural SSM block and sequence length-adaptive strategy are key innovations
- Achieves SOTA results on multiple point cloud tasks
- Ablation and visualization confirm effectiveness of each module